Inspiring Innovation and Leadership

Karatina University

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

BACHELOR OF COMPUTER SCIENCE, BACHELOR OF INFORMATION TECHNOLOGY, BACHELOR OF EDUCATION (SCIENCE) AND BACHELOR OF SCIENCE WITH EDUCATION

COM 333: DESIGN AND ANALYSIS OF ALGORITHMS

ASSIGNMENT 1 **DUE:** 19th December 2025

Instructions: Attempt all questions and submit via the link on ODeL.

1) Below are some concepts related to the analysis & Design of Algorithms:
   a) Binary tree **(2 Marks)**
   b) Backtracking **(2 Marks)**
   c) Dynamic programming **(2 Marks)**
   d) Greedy algorithms **(2 Marks)**
   e) 'NP' problems **(2 Marks)**

   Required: Using an example/illustration, **explain** each concept.

2) **Discuss** the below classical problems giving the **procedure** followed in solving each of them
   a) Travelling Salesman problem **(3 Marks)**
   b) knapsack problem **(3 Marks)**
   c) The tower of Hanoi **(3 Marks)**

3) **Write** *algorithms* to perform the following:
   a) Change a volume in Litres to Pints (1 Litre = 2.2 Pints) **(2 Marks)**
   b) Quicksort the set X = {12, 81, 30, 49, 53, 70, 66} in descending order **(3 Marks)**
   c) Find and display the **smallest** of four numbers. Assume the numbers are input one after the other from the keyboard. **(3 Marks)**

4) Describe the following properties of asymptotic complexities:
   a) Transitivity **(2 Marks)**
   b) Reflexivity **(2 Marks)**
   c) Symmetry **(2 Marks)**
   d) Complementarity **(2 Marks)**